

sTMS Interface Functional Specification

Temasek Polytechnic – Alumni System

Version 1.0

Document Version

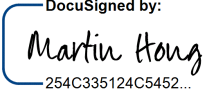
Version	Effective Date	Summary of Changes	Created/Updated By
0.1	25/07/2022	The initial version of sTMS – TP Alumni System interface functional specification.	AvePoint
0.2	02/08/2022	Revised base on TP feedback	AvePoint
1.0	04/08/2022	Finalized version for TP Alumni System Interface Specification	AvePoint

Document Acceptance Form

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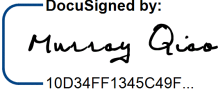
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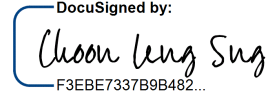
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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS shared Training Management System (sTMS) and TP Alumni System. This interface will be a 2-way interface between the system, what is sent back to TP from sTMS are considered as CET Alumni hence TP Alumni System will only send PET alumni details to sTMS.

The purpose of this interface is for TP alumni to enjoy alumni discounts when applying for specific courses in sTMS. It is required to push sTMS CET alumni information to TP Alumni System and TP Alumni System will sync back PET alumni information to sTMS.

1.2 Intended Audience

This technical document is used for system-to-system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

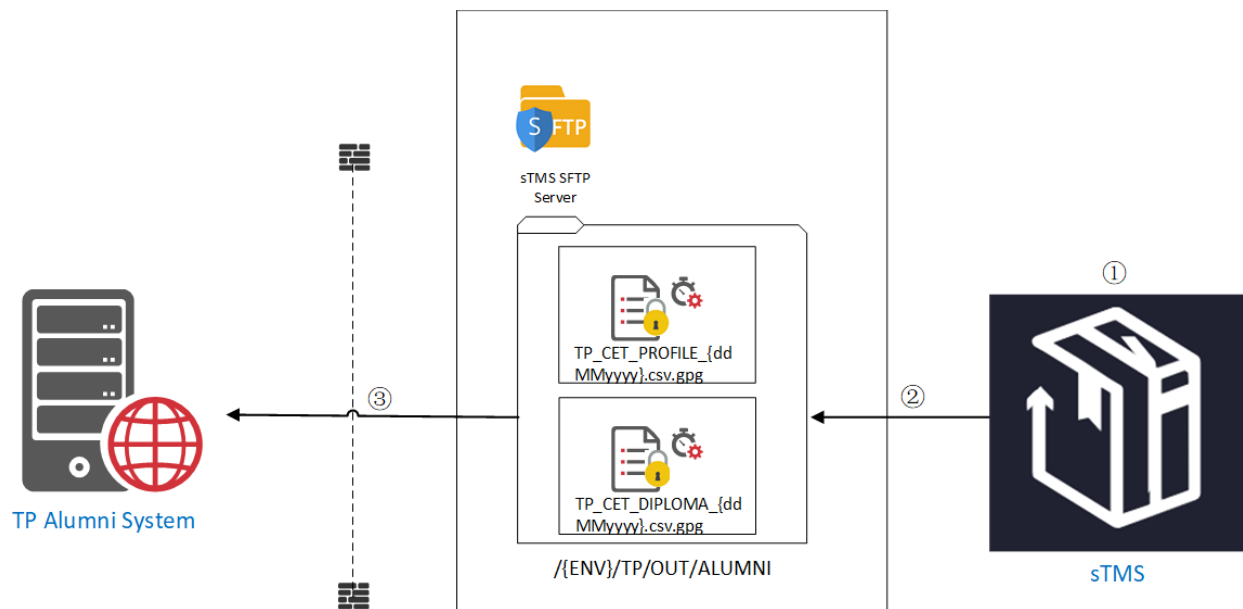
2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

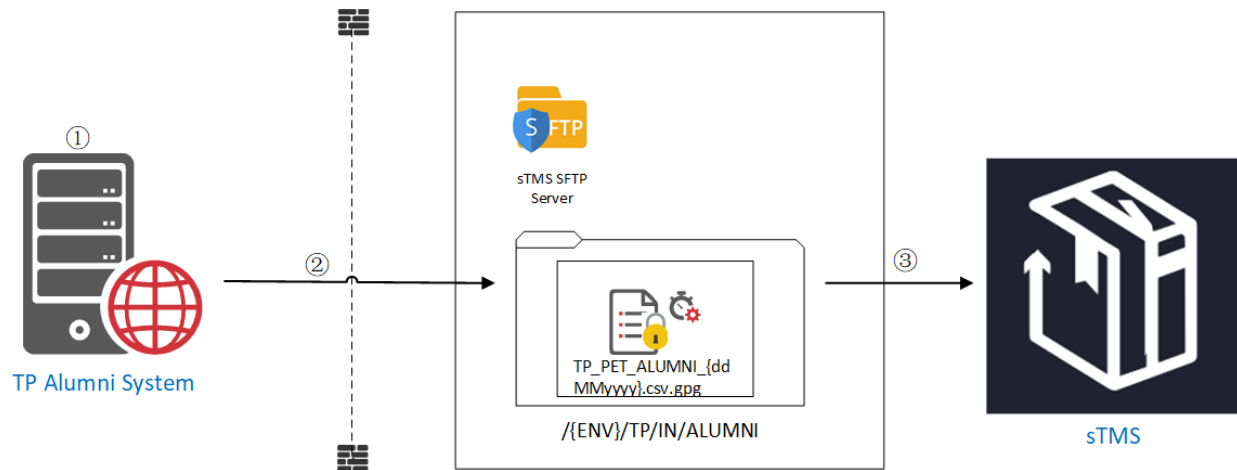
- **sTMS**: shared Training Management System.
- **sTMS SFTP Server**: A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **TP Alumni System**: A TP internal system that contain TP alumni details.

2.1.1 From sTMS to TP Alumni System



SN	Description
1	sTMS generates CSV files of TP CET alumni info on a pre-defined schedule base on system setting or manually triggered by CET users (File generation rule please refer to 2.3 Business Rule).
2	sTMS configures timer job to push CSV files into sTMS SFTP server after encryption with PGP protocol.
3	TP Alumni System pulls the TP CET graduate student information from sTMS SFTP server on demand.

2.1.2 From TP Alumni System to sTMS



SN	Description
1	User/job deposit data file into TP side SFTP server
2	TP side SFTP job pick up the file, encrypt with PGP and deposit into sTMS SFTP server
3	sTMS pulls the CSV file from sTMS SFTP server and stores the info as TP PET alumni.

2.2 Interface Design Summary

Output Interface File

Interface File Name	TP_CET_PROFILE_{ddMMyyyy}.csv.gpg TP_CET_DIPLOMA_{ddMMyyyy}.csv.gpg
Data Source System	shared Training Management System
Data Destination System	Temasek Polytechnic Alumni System
Method	SFTP
File Path	/[ENV]/TP/OUT/ALUMNI ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily
Timing	The cut off time for data will be 23:59:59 pm every day, timer job trigger timing is next day 01:00 am (TBC)
Folder Retention Policy	Files under the /[ENV]/TP/OUT/ALUMNI are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted. TP also can delete these files after pulling out

File Transfer Policy	sTMS shall transfer files if there are new records added/updated, otherwise no file will be generated and transfered
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not being found in sTMS SFTP server: ○ sTMS support team redeposits the file to sTMS SFTP server.

Input Interface File

Interface File Name	TP_PET_ALUMNI_{ddMMyyyy}.csv.gpg
Data Source System	Temasek Polytechnic Alumni System
Data Destination System	shared Training Management System
Method	SFTP
File Path	/{ENV}/TP/IN/ALUMNI ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily
Timing	The cut off time for data will be 23:59:59 pm every day, timer job trigger timing is next day 01:00 am (TBC)
Folder Retention Policy	Files under the /{ENV}/TP/IN/ALUMNI are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted.
File Transfer Policy	TP shall transfer files if there are new records added/updated, otherwise no file will be generated and transfered
Possible Exception and Exception Handling	Data error. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs and files to troubleshoot the cause of the failure (data error types are listed below). ○ Field exceeds max length allowed ○ Field is mandatory but missing • sTMS support team send email to TP team, TP redeposits the correct file in sTMS SFTP server.

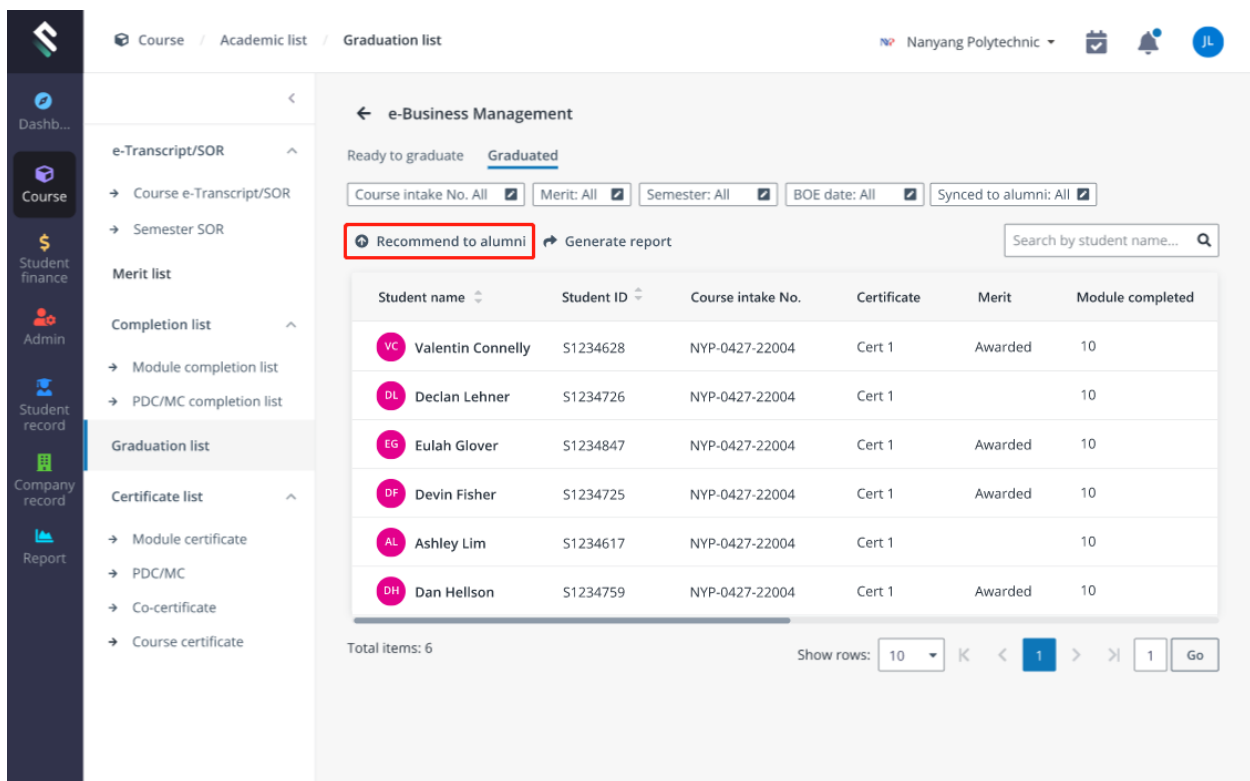
2.3 Business Rule

sTMS has automatic and manual configuration methods to send TP alumni info to TP Alumni System. If one method is successful, sTMS will not send repeated alumni info to TP again.

2.3.1 Manually Promote Alumni to TP Alumni System

The build-in configuration is set as a 'manual' action named **"Recommend to alumni"** to disseminate course completion details to TP Alumni System. By performing this action, it will trigger the system flow to send the graduated list information from sTMS to TP Alumni System. For graduation list details please refer to sTMS stage 3 functional specification: 9.4.6 Graduation List.

Refer to the clickable prototype here: [Graduation List](#)



The screenshot shows the 'e-Business Management' interface for 'Nanyang Polytechnic'. The left sidebar contains navigation links: Dashboard, Course, Student finance, Admin, Student record, Company record, and Report. The main content area is titled 'e-Business Management' and has tabs for 'Ready to graduate' and 'Graduated'. Under the 'Graduated' tab, there are filters for 'Course intake No.', 'Merit', 'Semester', 'BOE date', and 'Synced to alumni', all set to 'All'. A red box highlights the 'Recommend to alumni' button, which is next to a 'Generate report' button. Below these buttons is a table with the following data:

Student name	Student ID	Course intake No.	Certificate	Merit	Module completed
VC Valentin Connolly	S1234628	NYP-0427-22004	Cert 1	Awarded	10
DL Declan Lehner	S1234726	NYP-0427-22004	Cert 1		10
EG Eulah Glover	S1234847	NYP-0427-22004	Cert 1	Awarded	10
DF Devin Fisher	S1234725	NYP-0427-22004	Cert 1	Awarded	10
AL Ashley Lim	S1234617	NYP-0427-22004	Cert 1		10
DH Dan Hellson	S1234759	NYP-0427-22004	Cert 1	Awarded	10

At the bottom of the table, it says 'Total items: 6'. There is a pagination control showing 'Show rows: 10' and a page number '1'.

2.3.1 Automatically Promote Alumni to TP Alumni System

The system also allows TP to configure automated flows on top of the build-in manual mode, the configuration is also flexible down to the specific course level if such a configuration is required. For alumni configuration details please refer to sTMS stage 3 functional specification: 11. Alumni Management Interface.

Refer to the clickable prototype here: [Alumni Setting](#)



Dashb...

Course

Student finance

Admin

Student record

Company record

Report

Admin / Service management / Alumni setting

Role management

Account management

Master data

Form bank

Rule management

Service management

Email notification

Template

Document template

Digital certificate template

Certificate template

Transcript template

Signatures

Certificate image library

Admin settings

Announcement management

Student resource

Alumni management

Finance settings

Odd-cents write-off setting

Refund priority setting

Bank file setting

Student virtual card

User activity report

Status: All

Create

Edit

Activate

Name

Alumni Setting of Course type

Total items: 1

Create alumni setting

Name *

Description

Scope configuration

Select the scope to apply

☒ A combination of factors ☐ Specific courses

☒ Course category

Part-time Diploma (PTD)

Certificate course

☒ Course type

Academic Full Qualification (FQ)

SkillFuture Series

☒ Branch

SIRS

☒ School

School of business

Details

Choose a condition *

☐ On course completion date

☒ 10 Weeks after course completion date

☐ Finish % modules and are competent

☐ on result release date

☐ 10 Weeks after result release date

Cancel

Save

3. Interface Fields

3.1 CSV Files Interface Field

3.1.1 CET Alumni Information Interface Fields

Profile of Alumni

TP Field Name	sTMS Field Name	Description	Data Type	Remark
NRIC/FIN	NRICFIN	NRIC/FIN of alumni	String(\$nvarchar(20))	Sample Value: G1234567P
Passport No.	PassportNo	Passport number of alumni	String(\$nvarchar(20))	Sample Value: E123456789
Name	Name	Name of alumni	String(\$nvarchar(max))	Sample Value: Jesse Tan
Race	Race	Race of alumni	String(\$nvarchar(20))	Sample Value: Chinese
NSF Status	NSStatus	Status of alumni national service	String(\$nvarchar(200))	Value list: Complete NS, ORD date, Performing NS, Start date, Not applicable, Exempted.
Religion	Religious	Religious preference of alumni	String(\$nvarchar(200))	Value list: Buddhist, Christian, Catholic, Hindu, Jewish, Muslim, No Religion, Sikh, Taoist, Others.
Date of Birth	DateOfBirth	Date of birth of alumni	String(\$DD/MM/YYYY)	Sample value: 28/03/1992
Nationality	Nationality	Nationality of alumni	String(\$nvarchar(20))	Sample Value: Singapore
Gender	Gender	Gender of alumni	String(\$nvarchar(10))	Value list: Male, Female
Block/House No.	BlockNo	Block/House number	String(\$nvarchar(20))	Sample value: BLK123
Building Name	BuildingName	Building name	String(\$nvarchar(max))	Sample value: Building A
Street Name	StreetName	Street name	String(\$nvarchar(max))	Sample value: Street A
Floor No.	FloorNo	Floor number	String(\$nvarchar(10))	Sample value: 08
Unit No.	UnitNo	Unit number	String(\$nvarchar(10))	Sample value: 33
Postal code	PostalCode	Postal code of residential address	String(\$nvarchar(20))	Sample value: 307717
Foreign Address	AddressLine1+AddressLine2	Foreign Address of alumni	String(\$nvarchar(max))	

Country	CountryCode	Country of residential address	String(\$nvarchar(100))	Sample value: Singapore
Primary Email	Email	Personal email of alumni	String(\$nvarchar(100))	Sample Value: Jake@email.com
Mobile No.	MobileNo	Mobile number of alumni	String(\$nvarchar(20))	Sample value: 12345678
Employment Status	EmploymentStatus	Employment status of alumni	String(\$nvarchar(20))	Value list: Employed, Unemployed
Company	CompanyName	Company name of alumni	String(\$nvarchar(max))	Sample value: AxxxxP
Position	PositionHeld	Position held of alumni	String(\$nvarchar(100))	Sample value: Business Analyst

Diploma of Alumni

TP Field Name	sTMS Field Name	Description	Data Type	Remark
NRIC/Passport No.	NRICFIN	NRIC/FIN/Passport number of alumni	String(\$nvarchar(20))	Sample Value: G1234567P
Name	Name	Name of alumni	String(\$nvarchar(max))	Sample Value: Jesse Tan
TP Diploma	CertificateName	Certificate name of diploma	String(\$nvarchar(max))	Sample value: Diploma in Information technology
Course Code	CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
Course Name	CourseName	Course name	String(\$nvarchar(200))	Sample Value: Information Technology
School	SCHOOL	School name	String(\$nvarchar(32))	Sample Value: School of Business
Admission No.	StudentId	UAM common Id of alumni	String(\$nvarchar(11))	Sample Value: S123457
Course Type	CourseType	Course type	String(\$nvarchar(20))	Sample Value: Academic Full Qualification
Graduation Date	GraduationDate	Date of graduation	String(\$DD/MM/YYYY)	Sample value: 01/07/2022

3.1.2 PET Alumni Information Interface Fields

Name	Description	Data Type	Remark
NRIC/FIN	NRIC/FIN number of alumni	String(\$nvarchar(20))	Sample Value: G1234567P
Passport	Passport number of alumni	String(\$nvarchar(20))	Sample Value: X2243421F
Name	Name of alumni	String(\$nvarchar(max))	Sample Value: Jesse Tan

CourseCode	Refers to PET course code the TP alumni graduated from	String(\$nvarchar(50))	
CourseName	PET course name the TP alumni graduated from	String(\$nvarchar(50))	
GraduationYear	Refers to the graduation year related to the course	String(\$nvarchar(4))	Sample Value: 2020
StudentType	Store if this student is a PET or CET alumni.	String(\$nvarchar(10))	Default to PET

3.2 Sample File



TP_CET_PROFILE_25072022.csv



TP_PET_ALUMNI_25072022.csv



TP_CET_DIPLOMA_25072022.csv